Assignment 1 :Git & GitHub Fundamentals

Session 1 : Group A (Individual Lab Report)

Full Name: Chaimae TAHIR

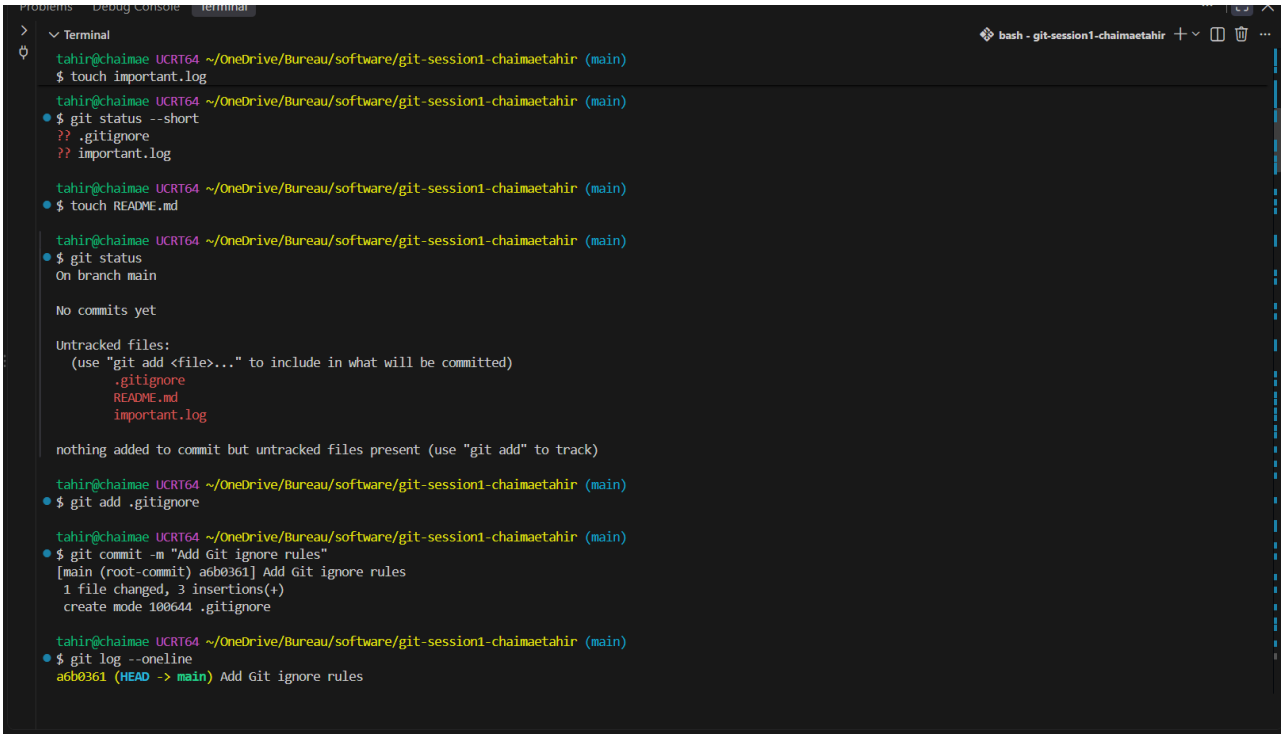
ID: 120024

GitHub Username: chaimaetahir

Repository: https://github.com/chaimaetahir/git-session1-chaimaetahir

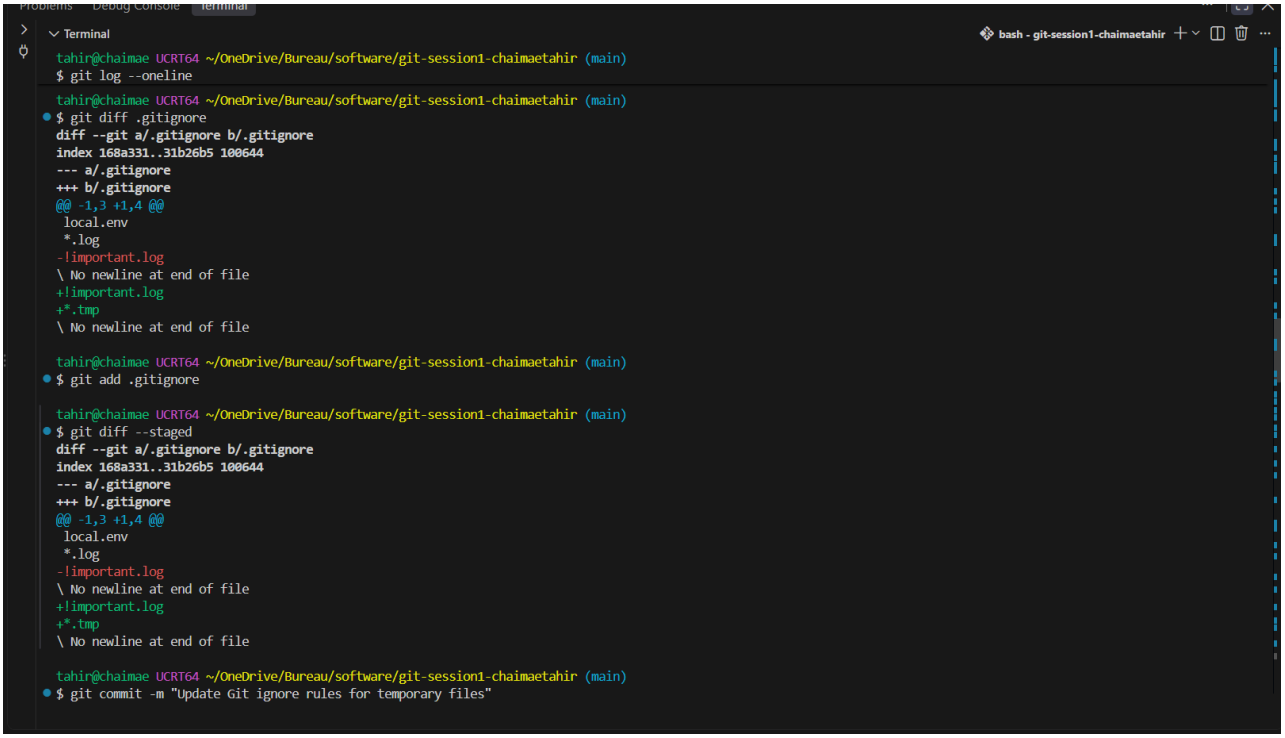
1. Repository Setup

I created the project git-session1-chaimaetahir in VS Code and initialized a Git repository on the main branch. My .gitignore excludes local.env, ordinary .log files and temporary .tmp files, but specifically allows important.log. I verified the ignore behavior with git status, then committed the configuration.



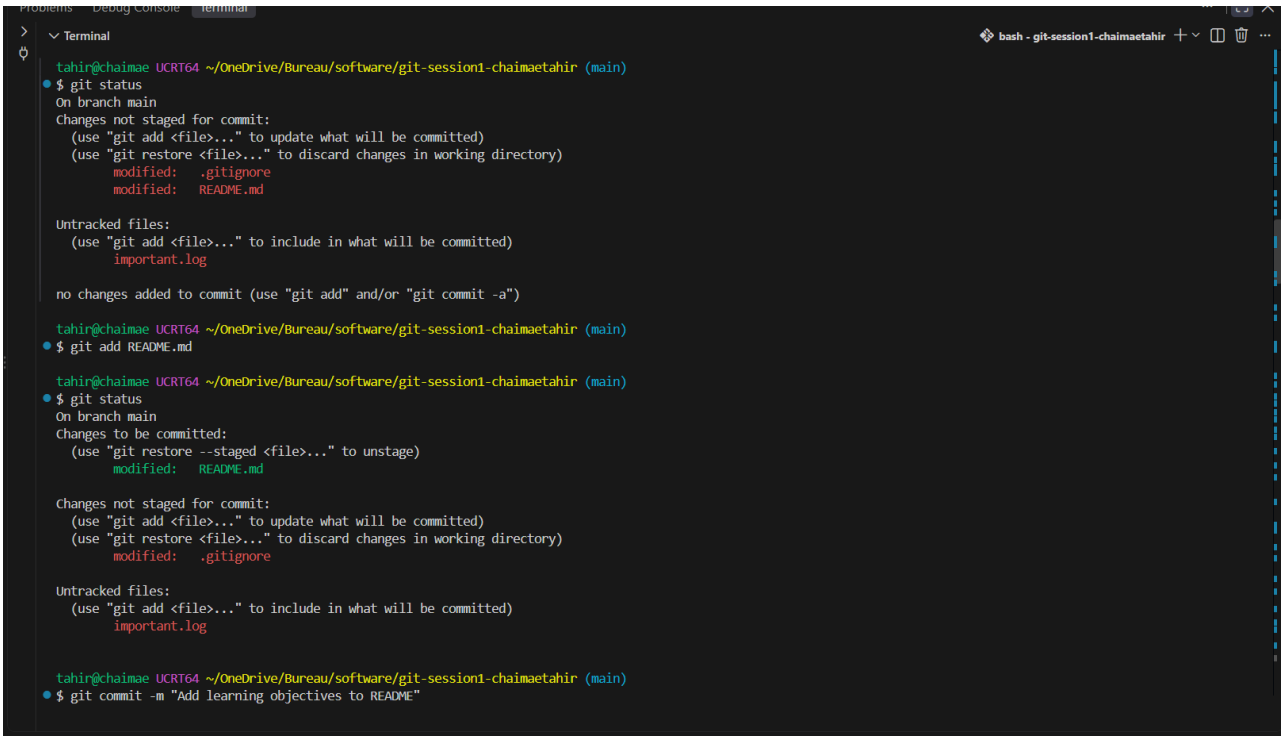
2. README & Commit History

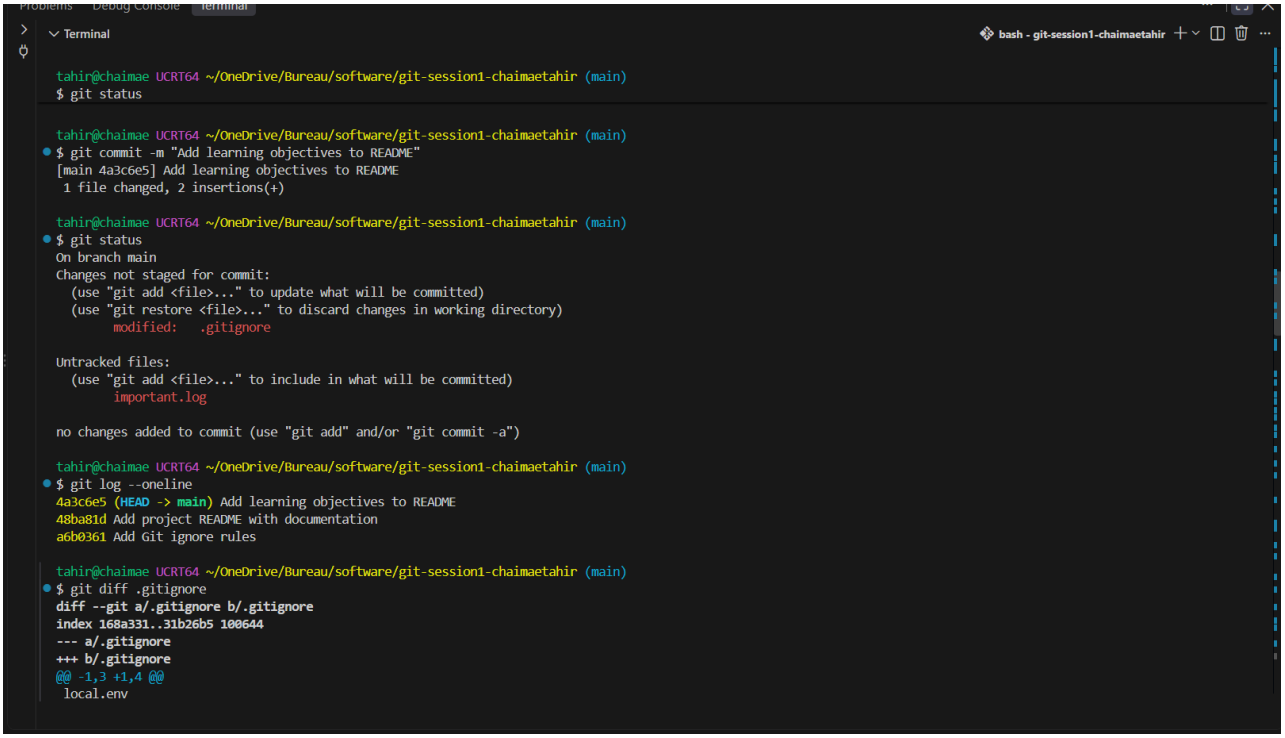
My README introduces the project, lists student information and technologies, and includes a command example, documentation link, and an image reference. I organized changes into focused commits, including ignore rules, README documentation, learning objectives, an About feature, and the important log file.



3. Selective Staging

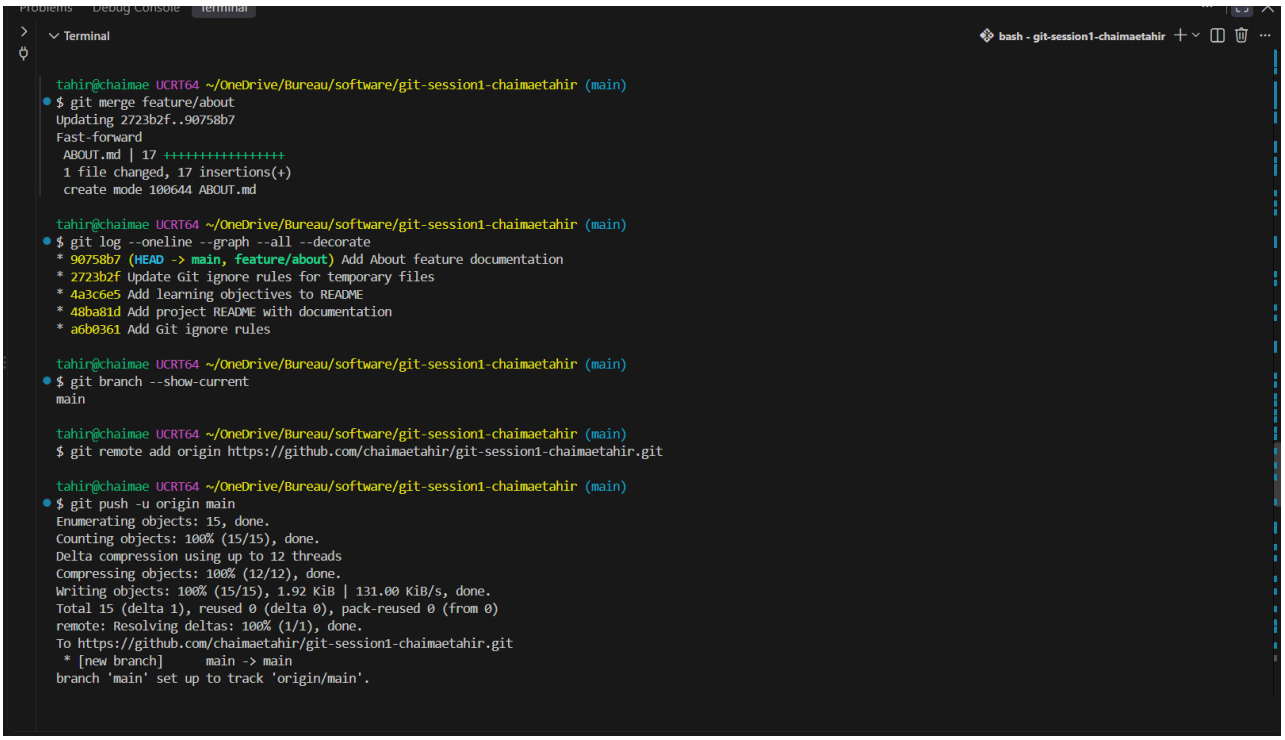
I edited README.md and .gitignore at the same time, staged only README.md using git add README.md, and committed the learning objectives. The .gitignore changes stayed unstaged until a later commit. I also compared changes with git diff and git diff --staged.





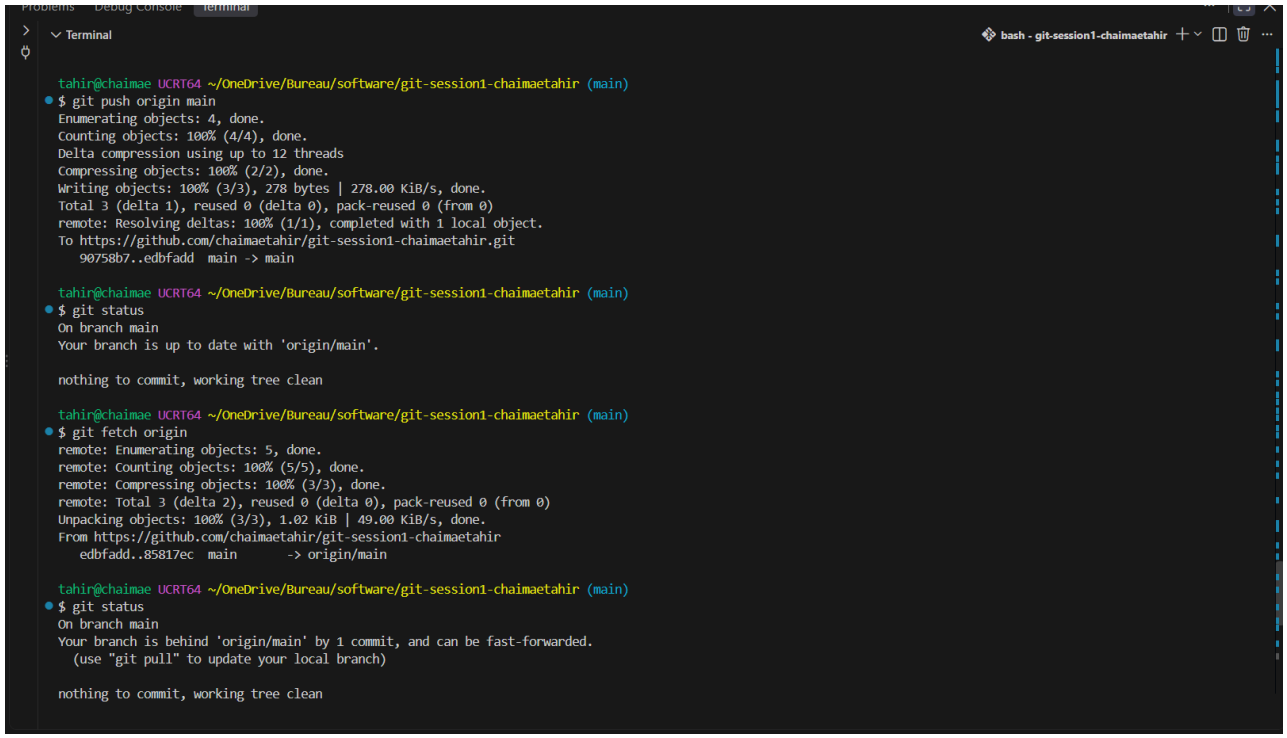
4. Branch & Merge

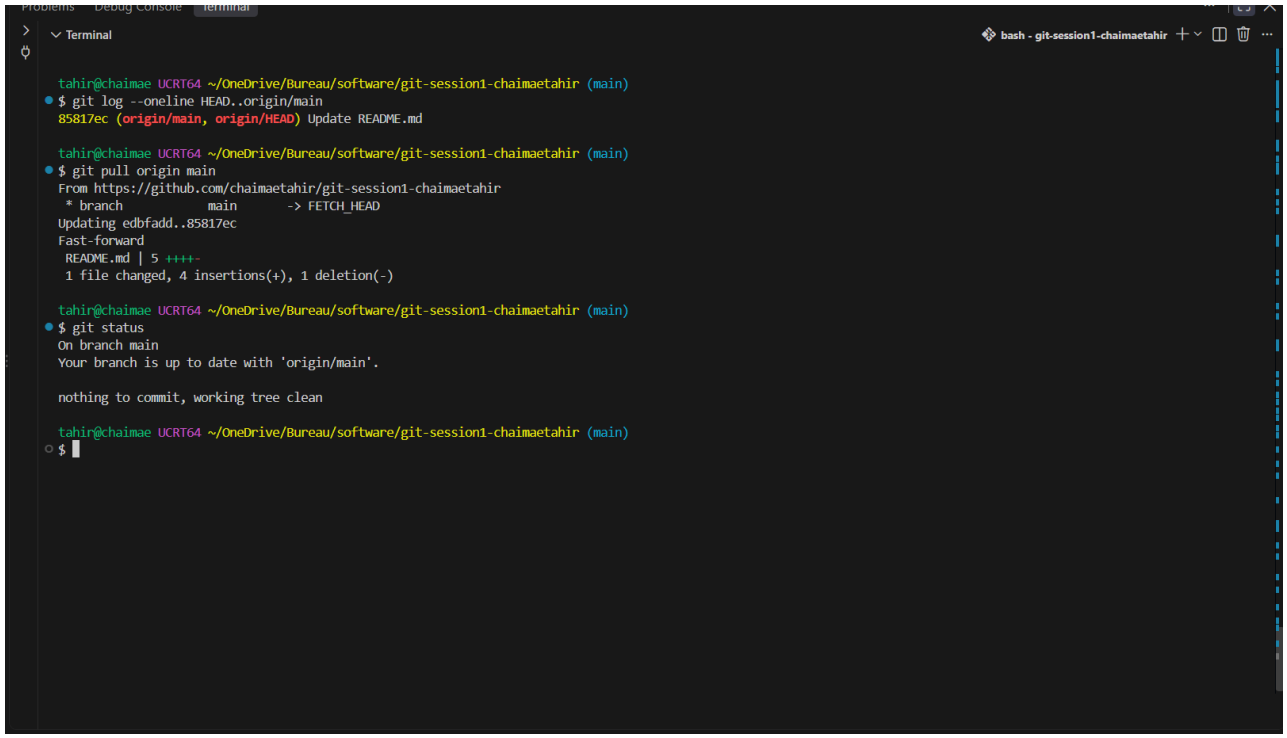
I created the feature/about branch using git switch -c feature/about, added ABOUT.md, and committed the new documentation on that branch. I returned to main and merged the branch using git merge feature/about. Git performed a fast-forward merge because main had not advanced separately.



5. Remote Changes

A commit records changes in my local repository, while push uploads local commits to GitHub. Fetch downloads remote history without changing local working files, whereas pull fetches and integrates those changes into the current branch. I edited README.md directly on GitHub, fetched the update, confirmed main was one commit behind origin/main, then pulled the change and checked that my working tree was clean.





6. Fair Use of AI

I completed the practical tasks of this assignment using VS Code, Git Bash, and GitHub. I used ChatGPT occasionally to clarify certain Git commands, understand concepts such as selective staging, branching, merging, fetch, and pull, and get help with some explanations and comments. I performed the commands myself, reviewed the terminal outputs, and verified the results using git status, git log, and screenshots. ChatGPT was mainly used as a supporting learning resource when I needed additional clarification.

7. Sources & Reflection

Sources: Assignment instructions and course materials, Git Documentation, GitHub Documentation, and ChatGPT for occasional explanations.

The most challenging part of this assignment was selective staging because I had to understand how to commit changes to one file while leaving another modified. Through practice, I became more familiar with managing changes in Git. If Git behaves unexpectedly, I would first check git status, then use git diff or git log to identify the issue.